Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 5: Position–Time Graphs — ANSWERS

The Basic 6 asked you which picture matched which motion. This sheet asks for numbers. One graph runs through Sections 1 and 2 — a walk to the shop and back. You take two points off it, put them through the gradient formula, and get a velocity in m/s. Section 3 is a new journey, and this time you draw the graph yourself before you read anything off it.

Section 1 — Gradient = velocity

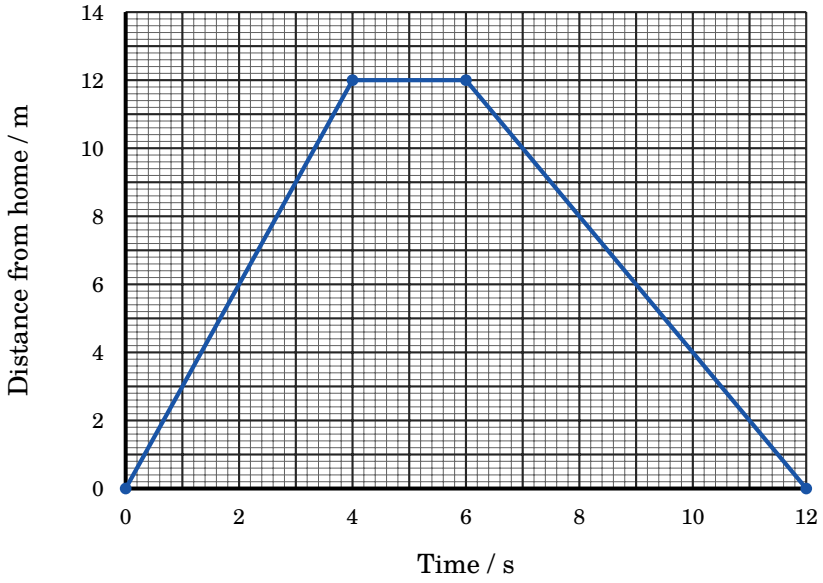
aP5-48, aP5-51

The gradient of a position–time graph is the velocity. It is the same gradient formula you used on straight lines:

$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

Pick two points on a straight section of the line and read off their coordinates. Here the y values are **distances** in metres and the x values are **times** in seconds, so the gradient comes out as a **velocity** in m/s.

Take the two points in the same order top and bottom — point 2 minus point 1, both times — and the sign looks after itself. A **flat** line gives a gradient of zero: at rest. A line sloping **back down** towards the time axis gives a **negative** gradient: moving back towards the start.



Worked example — copy this shape

The first 4 seconds. A student walks from home towards the shop. Read the two ends of that straight section off the graph:

point 1 = (0 s, 0 m) point 2 = (4 s, 12 m)

gradient $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{12 - 0}{4 - 0} = \dfrac{12}{4} =$ **3 m/s** (away from home)

1. What is the student's velocity between **4 s and 6 s**? **0 m/s**

Point 1 = (4, 12), point 2 = (6, 12), so gradient = (12 − 12) ÷ (6 − 4) = 0 ÷ 2 = 0. The student is standing still at the shop, 12 m from home.

2. What is the student's velocity between **6 s and 12 s**? Write down both points first. **−2 m/s**

Point 1 = (6, 12), point 2 = (12, 0), so gradient = (0 − 12) ÷ (12 − 6) = −12 ÷ 6 = −2 m/s. The minus sign means walking back towards home.

Check your answers against the shape of the line. The walk out is the *steeper* of the two sloping sections, so it should give the larger answer — 3 against 2. The walk home covers the same 12 m but takes longer, so it is both gentler and **negative**, because that section comes back down the page. The flat middle gives exactly zero.

Section 2 — Reading the journey off the graph

aP5-35..37, aP5-43, aP5-47

Still the walk to the shop. Every answer below is read off the same graph.

3. **How far from home** is the student at **10 s**? **4 m**

Read straight up from 10 s on the time axis to the line, then across: 4 m. They are on the way home and nearly back.

4. **How long** does the student spend standing still? **2 s**

From 4 s to 6 s — the flat section. 6 − 4 = 2 s.

5. **How far did the student actually walk**, start to finish? And what was their **displacement** at 12 s?

Distance walked **24 m** Displacement **0 m**

12 m out and 12 m back is 24 m of walking, but the student finishes exactly where they started, so the displacement is zero. A line that comes back down to the time axis is a journey that ended at home.

6. **Average speed** for the whole 12 seconds. **2 m/s**

Total distance ÷ total time = 24 ÷ 12 = 2 m/s. Note this uses the 24 m they walked, not the 0 m displacement — speed is about the ground covered.

Two different answers from one journey, and both are correct. The student walked 24 m but ended up 0 m from home. Distance counts every step; displacement only asks where you finished compared with where you began. Any time a position–time line returns to where it started, you have a journey with distance but no displacement.

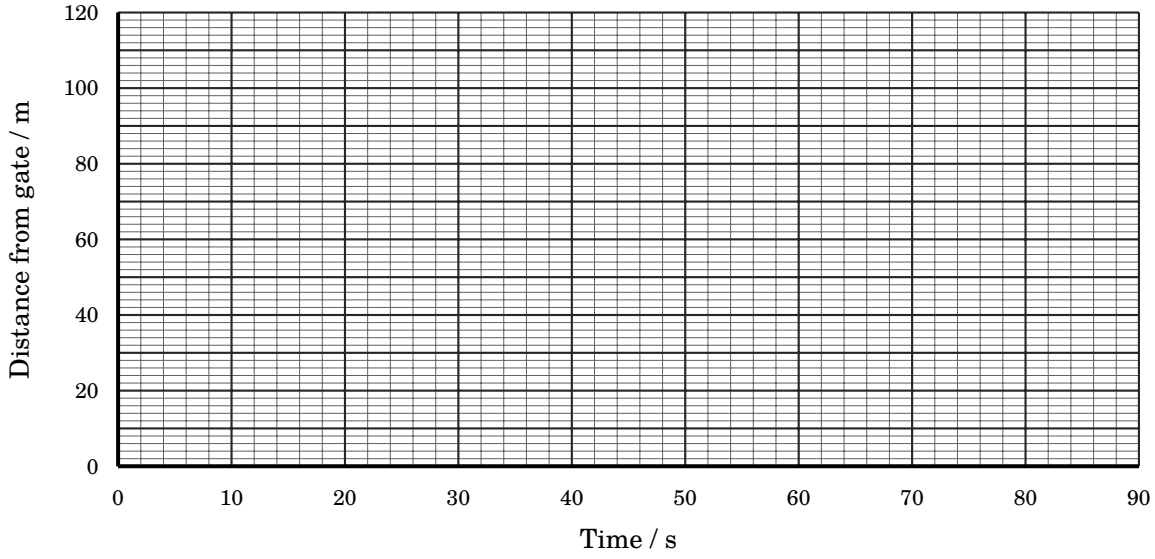
Section 3 — Your turn: the dog in the park

aP5-50, aP5-51, aP5-43

A dog is let off the lead at the park gate. It waits, runs off across the field, stops to investigate something, runs further still, stops again, then heads back towards the gate. Its distance from the gate was recorded every 10 seconds.

Time / s	0	10	20	30	40	50	60	70	80	90
Distance from gate / m	0	0	20	40	40	70	100	100	60	20

7. Plot the graph. Time on the horizontal axis, distance from the gate on the vertical. Plot every point from the table, then join them with a straight line between each pair. **Six straight sections** should appear — three where the dog is moving and three where it is not.



8. What is the dog's velocity between **10 s and 30 s**? **2 m/s**

Point 1 $= (10, 0)$*, point 2* $= (30, 40)$*, so gradient* $= (40 - 0) \div (30 - 10) = 40 \div 20 = 2$ *m/s.*

9. What is the dog's velocity between **40 s and 60 s**? **3 m/s**

Point 1 $= (40, 40)$*, point 2* $= (60, 100)$*, so gradient* $= (100 - 40) \div (60 - 40) = 60 \div 20 = 3$ *m/s. Steeper than the first run, so a bigger number.*

10. What is the dog's velocity between **70 s and 90 s**? **−4 m/s**

Point 1 $= (70, 100)$*, point 2* $= (90, 20)$*, so gradient* $= (20 - 100) \div (90 - 70) = -80 \div 20 = -4$ *m/s. Negative for heading back to the gate, and the fastest section of the whole journey.*

11. How much of the 90 seconds does the dog spend standing still? There is more than one flat section. **30 s**

Three flat sections: 0–10 s at the gate, 30–40 s, and 60–70 s. $10 + 10 + 10 = 30$ *s.*

12. How far did the dog run in total, and what was its **displacement** from the gate at 90 s?

Distance run **180 m** Displacement **20 m**

40 m, then 60 m further out, then 80 m back: $40 + 60 + 80 = 180$ *m of running. It finishes 20 m from the gate, so the displacement is 20 m — unlike the student in Section 2, this journey did not end where it started.*

13. Average speed for the whole 90 seconds. **2 m/s**

$180 \div 90 = 2\ m/s.$

Check the average against the graph. The dog's quickest section was 4 m/s and it spent 30 of the 90 seconds completely still, so an average of around 2 m/s is exactly what you would expect. An average speed should always come out smaller than the fastest section of the journey.